

@startuml

!include

https://raw.githubusercontent.com/plantuml-stdlib/C4-PlantUML/master/C4_Container.puml

LAYOUT_LEFT_RIGHT()

title AdScale Target Architecture (TO-BE)

Person(advertiser, "Advertiser")

Person_Ext(endUser, "End User")

System_Ext(dsp, "DSP Partner")

System_Ext(payment, "Payment Gateway")

System_Boundary(adscale, "AdScale Platform") {

Container(api, "API Gateway", "Kong / Nginx", "Routing, Authentication, Rate Limiting")

Container(adserver, "Ad Server Service", "Go", "Selects candidate ads")

Container(bidding, "Bidding Service", "Go/C++", "Auction calculation and winner selection")

Container(delivery, "Delivery Service", "Node.js", "Generates ad markup")

Container(events, "Event Collection Service", "Go", "Collects impressions and clicks")

Container(analytics, "Analytics Service", "Python", "Reporting and analytics")

Container(finance, "Financial Service", "Go", "Billing and balance management")

Container(campaign, "Campaign Service", "Go", "Campaign, targeting and creative management")

Container(notify, "Notification Service", "Go", "Email and system notifications")

Container(redis, "Redis Cache", "Redis", "Campaigns, targeting, bids cache")

Container(kafka, "Kafka", "Event Streaming Platform")

}

ContainerDb(campaignDb, "Campaign DB", "PostgreSQL", "Campaigns, creatives, targeting")

ContainerDb(financeDb, "Finance DB", "PostgreSQL", "Balances and financial transactions")

ContainerDb(analyticsDb, "Analytics DB", "ClickHouse", "Impressions, clicks, reporting")

' ===== External =====

Rel(dsp, api, "OpenRTB Requests")

Rel(advertiser, api, "Manage campaigns")

Rel(finance, payment, "Payments")

' ===== RTB Flow =====

Rel(endUser, delivery, "ads")

Rel(api, adserver, "Bid Request")

Rel(adserver, redis, "Read campaigns")

Rel(adserver, bidding, "Candidate ads")

Rel(bidding, redis, "Read bids")

Rel(bidding, delivery, "Auction winner")

Rel(delivery, events, "Impression / Click")

' ===== Campaign Management =====

Rel(api, campaign, "Campaign API")

Rel(campaign, campaignDb, "CRUD")

Rel(campaign, redis, "Update cache")

Rel(finance, campaign, "Budget updates")

' ===== Event Streaming =====

Rel(events, kafka, "Publish events")

Rel(campaign, kafka, "Campaign events")

Rel(kafka, analytics, "Consume events")

Rel(kafka, finance, "Billing events")

Rel(kafka, notify, "Notification events")

Rel(kafka, campaign, "Consume domain events")

' ===== Analytics =====

Rel(api, analytics, "Analytics API")

Rel(analytics, analyticsDb, "Read / Write")

' ===== Finance =====

Rel(api, finance, "Finance API")

Rel(finance, financeDb, "CRUD")

@enduml